Group 3 Project Proposal
An Empirical Study of Book-to-Market Portfolios in the U.S. Market (1982–2021)

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1) Introduction

Motivation

The book-to-market (B/M) ratio is a foundational metric in asset pricing. Historically, firms with high B/M ratios (value stocks) have outperformed those with low B/M ratios (growth stocks). This return differential has made B/M a central focus in both academic research and practical investing. With evolving markets and new data tools, it is critical to reassess the persistence and robustness of this effect.

Literature

The B/M ratio became central to asset pricing research after Fama and French (1992) demonstrated its power in explaining cross-sectional variation in stock returns. Building on this, Fama and French (1993) developed the three-factor model, introducing size (SMB) and value (HML) factors alongside market beta to improve upon the traditional CAPM. This model better captured systematic return patterns, especially those tied to firm fundamentals.

However, limitations remained. Fama and French (2015) extended their model into a five-factor version by adding profitability (RMW) and investment (CMA). While this improved model fit, it notably reduced the significance of the B/M effect, raising questions about its continued relevance.

In more recent work, Hou, Xue, and Zhang (2020) systematically replicated numerous pricing anomalies and confirmed that while B/M was once a strong predictor, its power weakened significantly after 2000. This reinforces the need to test whether the B/M premium still holds today. Meanwhile, Lakonishok, Shleifer, and Vishny (1994) offered a behavioral explanation for the value premium, arguing that mispricing driven by investor biases—rather than risk—may explain the higher returns observed in value stocks.

Our project builds on this literature by testing the B/M effect using updated CRSP and Compustat data, and comparing the results across multiple asset pricing models to assess whether the value premium remains significant in the modern era.

What We Hope to Find

We aim to replicate and test the performance of the B/M strategy using CRSP and Compustat data. We expect to observe a persistent value premium, especially in long-short portfolios. We will also explore whether the effect remains significant under expanded multi-factor models.

2) Methodology

Analysis Conducted

Using CRSP and Compustat data from 1982 to 2021, we calculate market capitalization as the product of absolute price and shares outstanding. Book equity is computed as: **seq + txditc − pstkrv**. We align book values with June market caps of the following year to compute B/M ratios. Each June, firms are sorted into five quintiles by B/M, and we track monthly returns from July of year *t* to June of year *t+1*. Portfolio returns are computed using **equal-weighted averages**.

We evaluate performance using:

1. Descriptive statistics: excess returns, standard deviation, skewness, and kurtosis
2. CAPM regressions: excess returns regressed on market excess return (Mkt-RF)
3. Fama-French three-factor regressions: excess returns regressed on Mkt-RF, SMB, and HML

Proposed Extension

We will expand to the Fama-French five-factor model by adding profitability (RMW) and investment

(CMA). To further enrich our analysis, we plan to incorporate momentum as a sixth factor (Carhart, 1997), enabling us to test whether value remains a distinct and significant return driver after accounting for broader firm characteristics.

3) Results and Interpretation

Univariate Performance

Average annualized returns increase steadily from 0.79% (Quintile 1) to 1.92% (Quintile 5). The long-short portfolio (Q5−Q1) generates a return of 1.16% with a t-statistic of 6.09, indicating high statistical significance ($p < 0.001$). These results are consistent with findings from both the academic literature and Ken French's website, validating the presence of a strong value premium.

CAPM Results

High B/M portfolios show significant alphas:

- **Q5**: 0.81% monthly (t = 3.68)
- **Q4**: 0.35% monthly (t = 2.41)
- **Long–Short**: 1.05% monthly (t = 5.36)
  Q1 shows a significantly negative alpha of −0.52% (t = −2.72). These results support the value premium hypothesis and show that CAPM alone does not fully capture return variations across B/M portfolios.

Fama-French 3-Factor Results

Adding SMB and HML improves model fit, yet significant alpha persists:

- **Q5**: 0.57% monthly (t = 3.65)
- **Q4**: 0.20% monthly (t = 2.13)
- **Q1**: −0.48% monthly (t = −4.12)
- **Long–Short**: 0.78% monthly (t = 6.08)
  R^2 values remain high, affirming the model's explanatory strength. These findings closely mirror those from the Ken French dataset and align with the empirical results in Fama and French (1993). However, they also confirm that even the three-factor model leaves room for unexplained variation, reinforcing the need for our proposed five- and six-factor extensions.

Comparison with Literature and Ken French's Website Data

Our results, generated using modified sample code and raw CRSP and Compustat data, are consistent with prior literature in demonstrating a persistent and significant value premium. This is in line with Fama and French (1993), whose three-factor model—incorporating market, size, and value factors—shows improved explanatory power over CAPM, as reflected in the higher R^2 values observed in our regressions. However, we cannot directly compare our results with Ken French's portfolio returns due to two key differences: French uses only NYSE firms to determine breakpoints, whereas our analysis includes all exchanges, and French's returns are value-weighted, while ours are equal-weighted.

Conclusion

- Excess Return: Portfolios 4 and 5 yield statistically significant excess returns.
- Findings vs. Literature: Our results confirm that high B/M portfolios earn higher average returns and that the long-short strategy delivers strong positive returns. The three-factor model, as expected from the literature, improves explanatory power over CAPM.
- Findings vs. Ken French Data: Although the directional results align with Ken French's data, direct comparison is not feasible due to differences in breakpoint methodology (NYSE-only vs. full market) and weighting scheme (value-weighted vs. equal-weighted).

References

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